Final Project Report:
How Music Taste Changes Over Time

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Abstract

Music has been a commonplace in daily life for ages, however, the way that people interpret music is very different. The 1960s were a very different time from present day and each person had very different tastes. In our case study, we are studying the billboards top 100 between the 1960s and Write a concise summary of your project here (usually 150-250 words). It should state the research question, briefly describe the data, mention the primary linear regression models used, and summarize the key findings and conclusions.

1 Introduction

Introduce your research question and the motivation behind your study. Why is this topic interesting or important? State the primary objectives of your regression analysis.

2 Data Description and Exploratory Data Analysis (EDA)

Describe the dataset you are using. Where did it come from? What are the observational units?

2.1 Variables

Briefly explain the response variable (Y) and the main predictor variables (X_i).

Let Y denote the target variable. The primary predictors are:

- X_1: Description of predictor 1.

- X_2: Description of predictor 2.

2.2 Exploratory Analysis

Present initial findings from your EDA. You can reference figures like scatterplots or correlation matrices here (e.g., "As seen in Figure 1...").

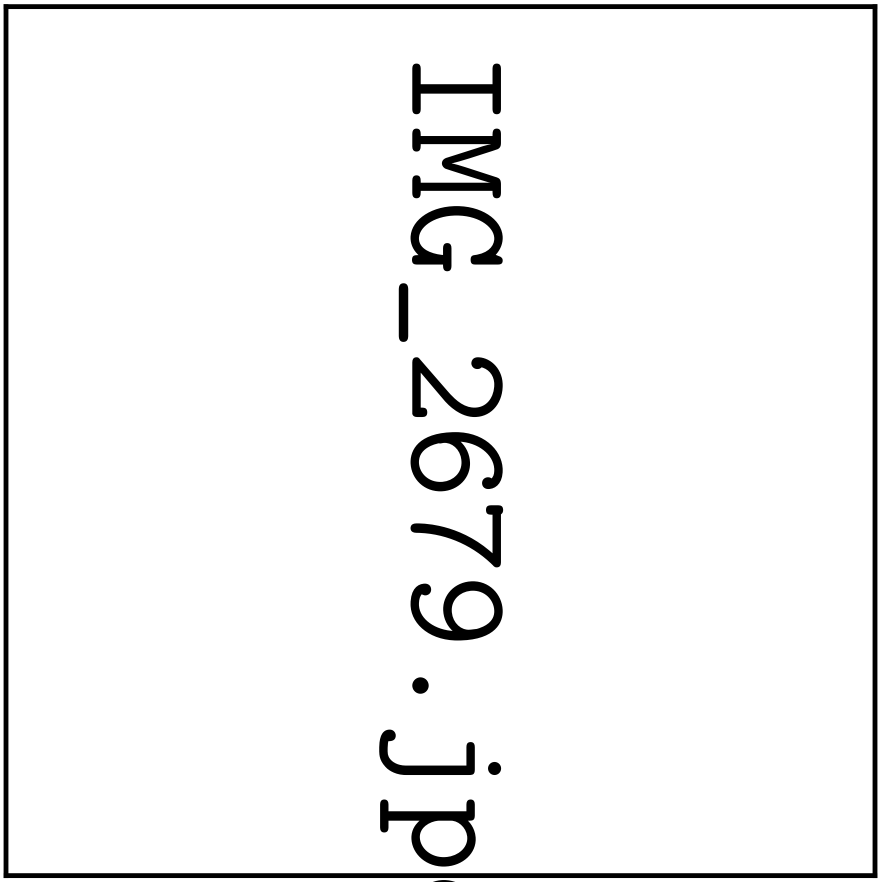

Figure 1: A descriptive caption explaining the plot.

3 Methodology and Model Building

Describe the models you fit to the data.

3.1 Proposed Model

We fit a multiple linear regression model of the form:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \epsilon_i \tag{1}$$

where $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Explain how you selected your variables (e.g., stepwise selection, theoretical justification, handling of multicollinearity). If you used transformations (like taking the log of the response), explain why.

4 Results and Diagnostics

Present the results of your final model.

4.1 Model Summary

Table 1 summarizes the estimated coefficients for our final model.

Table 1: Estimated Regression Coefficients and Significance

Predictor	**Estimate**	**Std. Error**	**t-value**	**Pr($> \|t\|$)**
(Intercept)	12.45	2.10	5.93	< 0.001
X_1	3.12	0.45	6.93	< 0.001
X_2	-1.05	0.30	-3.50	0.002

4.2 Model Diagnostics

Assess the assumptions of linear regression (linearity, independence, homoscedasticity, normality of residuals). You should include residual plots or QQ-plots here to justify that your model assumptions hold (or discuss how they are violated).

5 Discussion and Conclusion

Summarize the main takeaways from your model in the context of the original problem. What do the coefficients mean in the real world?

Acknowledge the limitations of your study (e.g., potential omitted variable bias, non-representative sample) and suggest areas for future work.

References

References

[1] Author Name. *Title of the Textbook*. Publisher, Year.

[2] Dataset Creator. "Title of the Dataset." *Source/URL*, Year.

A Appendix: R/Python Code

Paste your analysis code here. If you prefer, you can use the `verbatim` environment to format it as raw text:

```
# Example R Code
model <- lm(Y ~ X1 + X2, data = mydata)
summary(model)
plot(model)
```